

The conjectured generating function of OEIS A056328 is the one the entry already states

Adrian Perez Fontelles

Independent researcher

15 September 2026

Abstract

OEIS A056328 states one generating function as a fact and carries a second as an empirical conjecture. The conjecture is true, and no model of the sequence is needed to see it: the two expressions are the same rational function, so the conjectured form holds wherever the stated one does. The identity is exact, not a comparison of finitely many coefficients.

2020 Mathematics Subject Classification. 05A15, 11B37, 30B10.

1 The two statements

The entry's first terms are 0, 0, 0, 1, 6, 37, 183, 877, It states, as a fact and not as a conjecture,

$$F(x) = \frac{x^4 (3x^3 + x^2 - 2x + 1)}{-144x^7 + 228x^6 - 12x^5 - 121x^4 + 44x^3 + 12x^2 - 8x + 1},$$

and separately records the conjecture

Empirical g.f.: $-x^4(3x^3 + x^2 - 2x + 1) / ((x-1)(2x-1)(2x+1)(3x-1)(4x-1)(3x^2-1))$. - Colin Barker, Nov 25 2012

which is

$$G(x) = -\frac{x^4 (3x^3 + x^2 - 2x + 1)}{(x-1)(2x-1)(2x+1)(3x-1)(4x-1)(3x^2-1)}.$$

2 They are the same function

Both F and G are rational. Writing each over a common denominator and cancelling,

$$F(x) = \frac{-3x^7 - x^6 + 2x^5 - x^4}{144x^7 - 228x^6 + 12x^5 + 121x^4 - 44x^3 - 12x^2 + 8x - 1} = G(x),$$

so $F - G$ is identically zero as a rational function, and F and G have the same Taylor expansion at the origin wherever either converges.

Theorem 1. *The conjectured generating function of OEIS A056328 is correct.*

Proof. The entry asserts that F is the generating function of the sequence. By the identity above $G = F$, so G is that generating function as well. $\square$

3 What this rests on

It rests on the entry's own asserted generating function and on nothing else. That assertion is not taken on trust: F was expanded and its first 25 coefficients agree with every term the entry publishes, from offset 1 onward. The asserted line was read from the entry's factual formulas only – a generating function written inside a conjectural block would be the same conjecture stated twice and is not a premise.